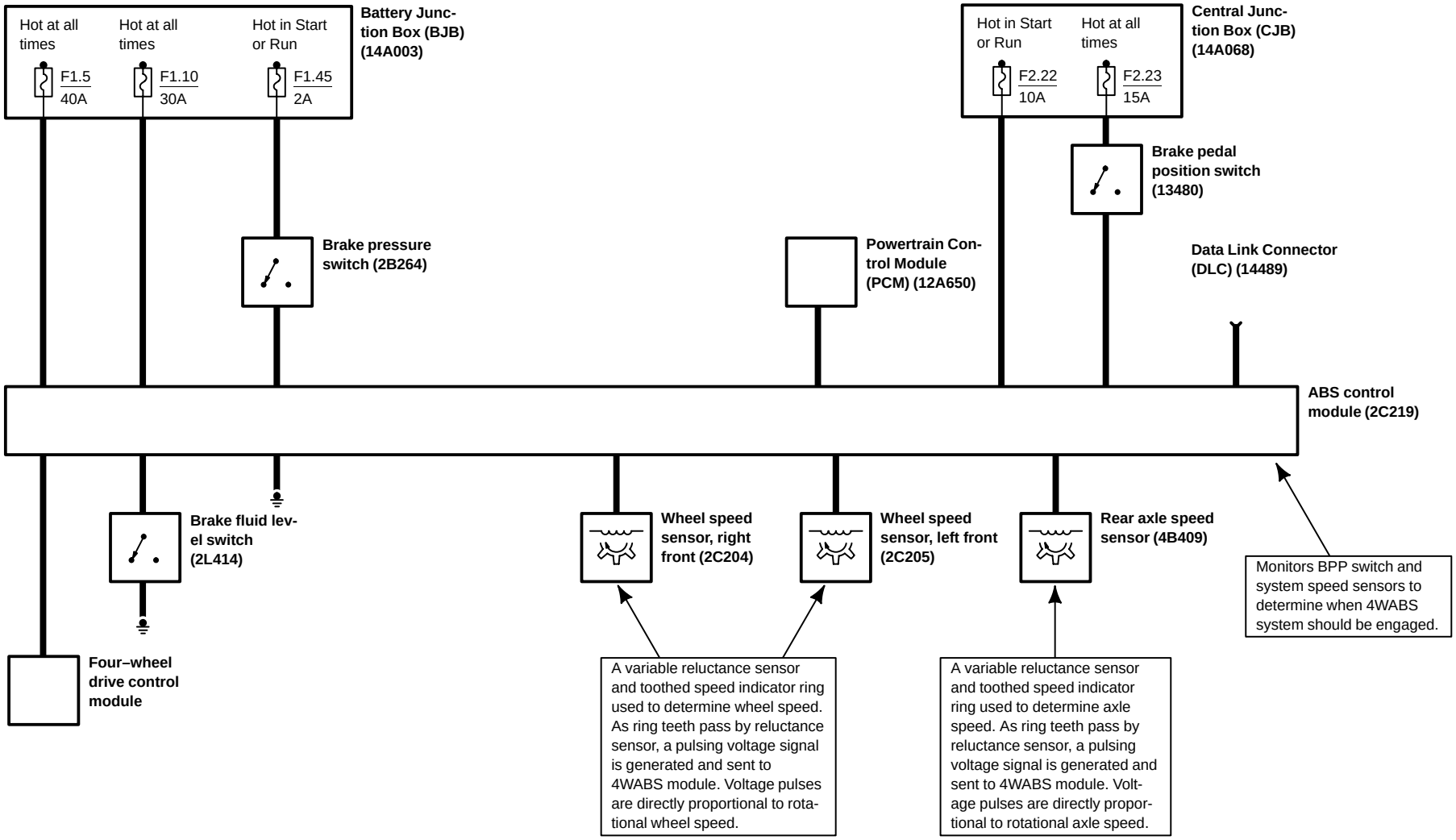without IVD, System overview